CSE 323: Operating System Design

Condition Variable & Semaphore

Salman Shamil



North South University (NSU)
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Original slides by Mathias Payer and Sanidhya Kashyap [EPFL]

Lecture Topics

- Condition Variables
- Producer-Consumer Problem
- Semaphores
- Signaling through condition Variables and Semaphores
- Concurrency Bugs

This slide deck covers chapters 30 and 31 in OSTEP.

Condition Variables (CV)

In concurrent programming, a common scenario is one thread waiting for another thread to complete an action.

```
1  bool done = false;
2
3  /* called in the child to signal termination */
4  void thr_exit() {
5   done = true;
6  }
7  /* called in the parent to wait for a child thread */
8  void thr_join() {
9    while (!done);
10 }
```

Condition Variables (CV)

- Locks enable mutual exclusion of a shared region.
  - Unfortunately they are oblivious to ordering
- Waiting and signaling (i.e., T2 waits until T1 completes a given task) could be implemented by spinning until the value changes

Condition Variables (CV)

- Locks enable mutual exclusion of a shared region.
  - Unfortunately they are oblivious to ordering
- Waiting and signaling (i.e., T2 waits until T1 completes a given task) could be implemented by spinning until the value changes
- But spinning is incredibly *inefficient*
- New synchronization primitive: ***condition variables***

Condition Variables (CV)

- A CV allows:
  - A thread to wait for a condition
  - Another thread signals the waiting thread
- Implement CV using queues

Condition Variables (CV)

- A CV allows:
  - A thread to wait for a condition
  - Another thread signals the waiting thread
- Implement CV using queues
- API: `wait`, `signal` or `broadcast`
  - `wait`: wait until a condition is satisfied
  - `signal`: wake up one waiting thread
  - `broadcast`: wake up all waiting threads
- On Linux, `pthreads` provides CV implementation

Signal parent that child has exited

```
1  bool done = false;
2  pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
3  pthread_cond_t c = PTHREAD_COND_INITIALIZER;
4  /* called in the child to signal termination */
5  void thr_exit() {
6    pthread_mutex_lock(&m);
7    done = true;
8    pthread_cond_signal(&c);
9    pthread_mutex_unlock(&m);
10  }
11 /* called in the parent to wait for a child thread */
12 void thr_join() {
13  pthread_mutex_lock(&m);
14  while (!done)
15    pthread_cond_wait(&c, &m);
16  pthread_mutex_unlock(&m);
17 }
```

Signal parent that child has exited (2)

- `pthread_cond_wait(pthread_cond_t *c, pthread_mutex_t *m)`
  - Assume mutex `m` is held; *atomically* unlock mutex when waiting, retake it when waking up
- Question: Why do we need to check a condition before sleeping?

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- Question: Why can't we use `if` when waiting?

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- Thread may have already exited, i.e., no need to wait
  - Principle: Check the condition before sleeping
- Question: Why can't we use `if` when waiting?
- There can be multiple threads to wait on the same CV. Race Condition!
  - Principle: `while` instead of `if` when waiting (more on this later...)

Signal parent that child has exited (3)

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Signal parent that child has exited (3)

- Question: Why do we need to proctect `done` with mutex `m`?
- Mutex `m` allows one thread to access `done` for protecting against missed updates
  - Parent reads `done == false` but is interrupted
  - Child sets `done = true` and signals but no one is waiting
  - Parent continues and goes to sleep (forever)
- Lock is therefore required for wait/signal synchronization

Producer/Consumer Problem

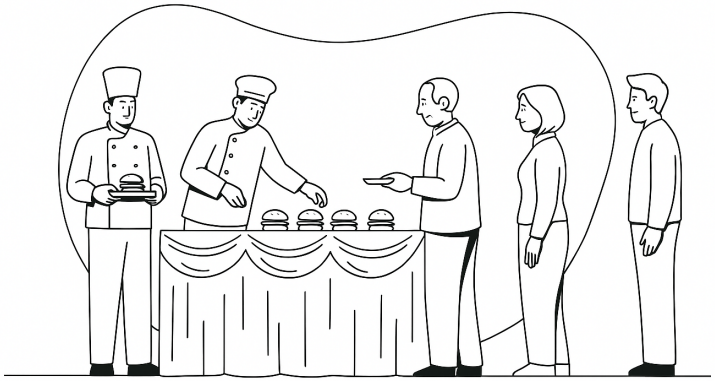


Figure 1: Producer-Consumer/Bounded Buffer Problem

- Producer/consumer is a common programming pattern
- For example: map (producers) / reduce (consumer)
- For example: a concurrent database (consumers) handling parallel requests from clients (producers)
  - Clients produce new requests (encoded in a queue)
  - Handlers consume these requests (popping from the queue)

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- One or more consumers process items from buffer
- Need synchronization for buffer
  - Want concurrent production and consumption
  - Use as many cores as available
  - Minimize access time to shared data structure
- Strategy: use CV to synchronize
  - Make producers wait if buffer is full
  - Make consumers wait if buffer is empty (nothing to consume)

Solving Producer/Consumer Problem

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```
int buffer;
int count = 0; // initially empty

void put(int value) {
    assert(count == 0);
    count = 1;
    buffer = value;
}

int get() {
    assert(count == 1);
    count = 0;
    return buffer;
}
```

```
void *producer(void *arg) {
  int i;
  int loops = (int) arg;
  for (i = 0; i < loops; i++) {
    put(i);
  }
}

void *consumer(void *arg) {
  while (1) {
    int tmp = get();
    printf("%d\n", tmp);
  }
}
```

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- **Problems with this solution**
  - Critical sections in put() and get(). **Use locks...**

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```

- **Problems with this solution**
  - Critical sections in `put()` and `get()`. **Use locks...**
- Producer-Consumer dependency for fetching. **Needs CV!**

Solving Producer/Consumer Problem

```
cond_t cond;
mutex_t mutex;

void *producer(void *arg) {
  int i;
  int loops = (int) arg;
  for (i = 0; i < loops; i++) {
    Pthread_mutex_lock(&mutex);
    if (count == 1)
      Pthread_cond_wait(&cond, &mutex);
    put(i);
    Pthread_cond_signal(&cond);
    Pthread_mutex_unlock(&mutex);
  }
}
```

```
void *consumer(void *arg) {
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  int loops = (int) arg;
  for (i = 0; i < loops; i++) {
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    if (count == 0)
      Pthread_cond_wait(&cond, &mutex);
    int tmp = get();
    Pthread_cond_signal(&cond);
    Pthread_mutex_unlock(&mutex);
    printf("%d\n", tmp);
  }
}
```

Does it work?

- Fine for single producer and single consumer.
- Change the setup to accommodate multiple producers and/or multiple consumers. How about now?

Solving Producer/Consumer Problem (2)

- **Setup:**
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- Consider a consumer thread (C1) is waiting for an item
- What if a second consumer thread (C2) sneaks in just after an item is produced? ... skipping the `wait()` call.
- Producer's `signal()` wakes C1 up, but C2 already fetched the item!
- Solution: Use `while` instead of `if` to recheck upon waking up.

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- What if a second consumer thread (C2) sneaks in just after an item is produced? ... skipping the `wait()` call.
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Producers and consumers both waiting on the same CV.

- Two consumers C1 and C2 runs and sleeps by calling `wait()`.
- Producer runs and `signal()` wakes up C1 (or C2).
- After consuming the item C1 can wake up producer again.
- But what if C1's `signal()` wakes up C2 instead?
- Solution: Use separate conditions for directed signaling.

Solving Producer/Consumer Problem (2)

```
cond_t empty, full;
mutex_t mutex;

void *producer(void *arg) {
  int i;
  int loops = (int) arg;
  for (i = 0; i < loops; i++) {
    Pthread_mutex_lock(&mutex);
    while (count == 1)
      Pthread_cond_wait(&empty, &mutex);
    put(i);
    Pthread_cond_signal(&full);
    Pthread_mutex_unlock(&mutex);
  }
}
```

```
void *consumer(void *arg) {
  int i;
  int loops = (int) arg;
  for (i = 0; i < loops; i++) {
    Pthread_mutex_lock(&mutex);
    while (count == 0)
      Pthread_cond_wait(&full, &mutex);
    int tmp = get();
    Pthread_cond_signal(&empty);
    Pthread_mutex_unlock(&mutex);
    printf("%d\n", tmp);
  }
}
```

Producer/Consumer Buffer with Multiple Slots

```c
int buffer[MAX];
int fill_ptr = 0;
int use_ptr = 0;
int count = 0;

void put(int value) {
  buffer[fill_ptr] = value;
  fill_ptr = (fill_ptr + 1) % MAX;
  count++;
}

int get() {
  int tmp = buffer[use_ptr];
  use_ptr = (use_ptr + 1) % MAX;
  count--;
  return tmp;
}


cond_t empty, fill;
mutex_t mutex;
```

```c
void *producer(void *arg) {
  int i;
  for (i = 0; i < loops; i++) {
    Pthread_mutex_lock(&mutex);
    while (count == MAX)
      Pthread_cond_wait(&empty, &mutex);
    put(i);
    Pthread_cond_signal(&fill);
    Pthread_mutex_unlock(&mutex);
  }
}

void *consumer(void *arg) {
  int i;
  for (i = 0; i < loops; i++) {
    Pthread_mutex_lock(&mutex);
    while (count == 0)
      Pthread_cond_wait(&fill, &mutex);
    int tmp = get();
    Pthread_cond_signal(&empty);
    Pthread_mutex_unlock(&mutex);
    printf("%d\n", tmp);
  }
}
```

Semaphore

- A semaphore extends a CV with an integer as internal state
- `int sem_init(sem_t *sem, unsigned int value)`: creates a new semaphore with `value` slots
- `int sem_wait(sem_t *sem)`: waits until the semaphore has at least one slot, decrements the number of slots
- `int sem_post(sem_t *sem)`: increments the semaphore (and wakes one waiting thread)
- `int sem_destroy(sem_t *sem)`: destroys the semaphore and releases any waiting threads

Producer/Consumer: Use Semaphores!

```
sem_t csem, psem;

/* BUFSIZE items are available for producer to create */
sem_init(&psem, 0, BUFSIZE);

/* 0 items are available for consumer */
sem_init(&csem, 0, 0);
```

Producer: Semaphores

```
1  void put(unsigned int val) {
2   /* we wait until there is buffer space available */
3   sem_wait(&psem);
4
5   /* store element in buffer */
6   buffer[ppos] = val;
7   ppos = (ppos + 1) % BUFSIZE;
8
9   /* notify consumer that data is available */
10  sem_post(&csem);
11 }
```

Consumer: Semaphores

```
1  unsigned int get() {
2   /* wait until data is produced */
3   sem_wait(&csem);
4
5   /* consumer entry */
6   unsigned long val = buffer[cpos];
7   cpos = (cpos + 1) % BUFSIZE;
8
9   /* notify producer that a space has freed up */
10  sem_post(&psem);
11  return val;
12 }
```

Producer/Consumer: Remaining Issues?

- We now synchronize between consumers and producers
  - Producer waits until buffer space is available
  - Consumer waits until data is ready

Producer/Consumer: Remaining Issues?

- We now synchronize between consumers and producers
  - Producer waits until buffer space is available
  - Consumer waits until data is ready
- How would you handle multiple producers/consumers?
  - Currently no synchronization between producers (or consumers)

Multiple Producers: Use Locking!

```
/* mutex handling mutual exclusive access to ppos */
1  pthread_mutex_t pmutex = PTHREAD_MUTEX_INITIALIZER;
2
3  void put(unsigned int val) {
4   unsigned int mypos;
5   /* we wait until there is buffer space available */
6   sem_wait(&psem);
7   /* ppos is shared between all producers */
8   pthread_mutex_lock(&pmutex);
9   mypos = ppos;
10  ppos = (ppos + 1) % BUFSIZE;
11  /* store information in buffer */
12  buffer[mypos] = val;
13  pthread_mutex_unlock(&pmutex);
14  sem_post(&csem);
15 }
```

Semaphores/Spin Locks/CVs are interchangeable

- Each is implementable through a combination of the others
- Depending on the use-case one is faster than the other
  - How often is the critical section executed?
  - How many threads compete for a critical section?
  - How long is the lock taken?

Implementing a Mutex with a Semaphore

```
1 sem_t sem;
2 sem_init(&sem, 1);
3
4 sem_wait(&sem);
5 ... // critical section
6 sem_post(&sem);
```

Implementing a Semaphore with CV/Locks

```
1  typedef struct {
2   int value;                  // sem value
3   pthread_mutex_t lock; // access to sem
4   pthread_cond_t cond;  // wait queue
5  } sem_t;
6
7  void sem_init(sem_t *s, int val) {
8   s->value = val;
9   pthread_mutex_init(&(s->lock), NULL);
10  pthread_cond_init(&(s->cond), NULL);
11 }
```

Implementing a Semaphore with CV/Locks

```
1   void sem_wait(sem_t *s) {
2     pthread_mutex_lock(&(s->lock));
3     while (s->value <= 0)
4         pthread_cond_wait(&(s->cond), &(s->lock));
5     s->value--;
6     pthread_mutex_unlock(&(s->lock));
7   }
8
9   void sem_post(sem_t *s) {
10    pthread_mutex_lock(&(s->lock));
11    s->value++;
12    pthread_cond_signal(&(s->cond));
13    pthread_mutex_unlock(&(s->lock));
14  }
```

Reader/Writer Locks

- A single (exclusive) writer, multiple (N) concurrent readers
- Implement using two semaphores: `lock` for the data structure, `wlock` for the writer
  - Both semaphores initialized with (1)
  - Writer only waits/posts on `wlock` when acquiring/releasing
  - Reader waits on `lock`, increments/decrements reader count
  - If number of `readers`==0, must wait/post on `wlock`

Reader/Writer Locks

```
1  void rwlock_acquire_readlock(rwlock_t *rw) {
2   sem_wait(&rw->lock);
3   rw->readers++;
4   if (rw->readers == 1)
5      sem_wait(&rw->wlock); // first r, also grab wlock
6   sem_post(&rw->lock);
7  }
8
9  void rwlock_release_readlock(rwlock_t *rw) {
10  sem_wait(&rw->lock);
11  rw->readers--;
13  if (rw->readers == 0)
14    sem_post(&rw->wlock); // last r, also release wlock
15  sem_post(&rw->lock);
16 }
```

- Full Code in Textbook (Page 12)

Summary

- Spin lock, CV, and semaphore synchronize multiple threads
  - Spin lock: atomic access, no ordering, spinning
  - Condition variable: atomic access, queue, OS primitive
  - Semaphore: shared access to critical section with (int) state
- All three primitives are equally powerful
  - Each primitive can be used to implement both other primitives
  - Performance may differ!